

Pulmonary Vascular Fractal Dimension on Non-Contrast Planning CT as a Potential Imaging Biomarker for Early Radiation-Induced Lung Injury: A Multi-Scale Structural Analysis

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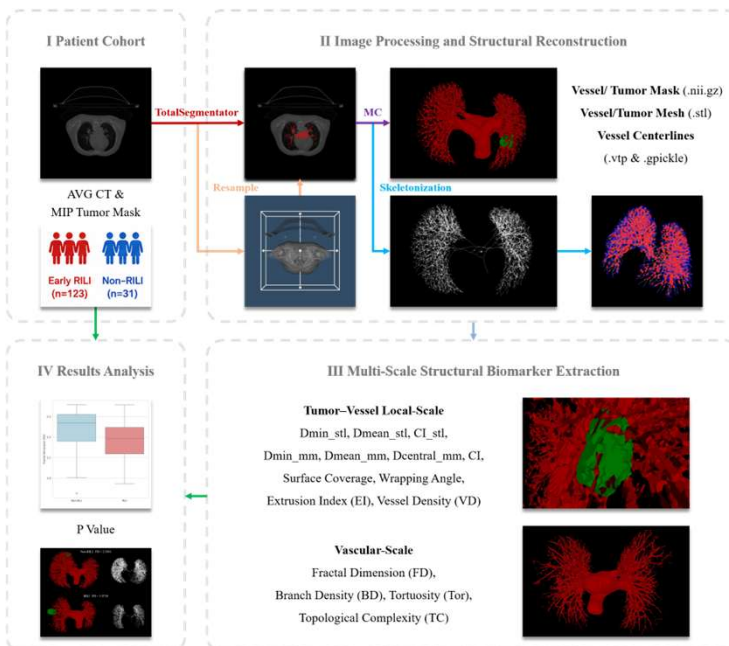
INTRODUCTION

- Lung cancer is a leading cause of cancer cases and deaths worldwide; ~60% of patients need ≥ 1 radiotherapy course after diagnosis.
 - Existing QIBA studies (dosimetrics/radiomics/ML) focus on whole-lung scale, but radiation-induced vascular damage is an established biomarker for RILI and prognosis.
- We aimed to explore vascular-scale structural biomarkers, bridging the gap between whole-lung and pulmonary vascular scales.



METHOD

This retrospective, single-center, ethics-approved study included 154 primary lung cancer patients receiving RT (2018–2025) with pre-RT non-contrast AVG CT, defined RT fields, and clinical data, divided into early RILI (n = 123) and non-RILI (n = 31; RILI 3–6 mo. post-RT).

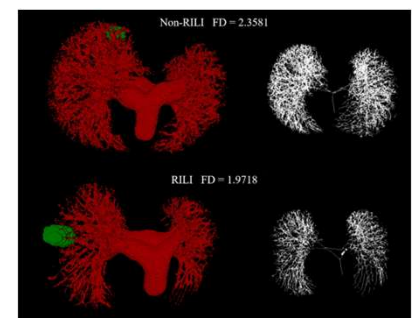
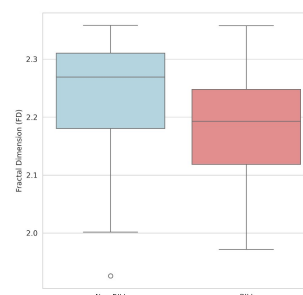


- ✓ Pre-RT CT processing used TotalSegmentator for pulmonary vessel segmentation, MC for tumor/vessel surface meshes, iterative thinning for 3D skeletonization, and 26-neighbor mapping to build an undirected graph $G(V,E)$.
- ✓ Fifteen candidate structural biomarkers were identified, including vascular-scale and tumor–vessel local-scale features.

RESULTS

Feature	Scale	Median Non-RILI*	Median RILI*	P-value	Cohen's d	Padj.*
FD	V	2.2691	2.1925	0.0055	-0.3949	0.0826
Dmin_stl	T-V	-7.348	-9.9499	0.0185	-0.2036	0.1391
Surface coverage	T-V	0.3219	0.2535	0.0789	-0.4459	0.2482
CI	T-V	0.0734	0.0591	0.1087	-0.4768	0.2482
EI	T-V	0.1928	0.1611	0.1107	-0.3403	0.2482
Dmean_mm	T-V	4.7763	5.5069	0.1137	0.2664	0.2482
Tor	V	1.1156	1.1227	0.1158	0.3975	0.2482
VD	T-V	0.0795	0.0633	0.2289	-0.2137	0.4293
Dmin_mm	T-V	1.1719	1.1719	0.2755	-0.2490	0.4591
CI_stl	T-V	0.5607	0.5405	0.3349	-0.2378	0.5024
BD	V	0.2176	0.2018	0.5077	-0.0146	0.6924
Dcentral_mm	T-V	87.0268	93.5312	0.5671	0.0481	0.7089
Wrapping_angle	T-V	77.1616	83.5994	0.6329	0.0574	0.7303
Dmean_stl	T-V	0.9782	0.9425	0.9461	-0.0322	0.9605
TC	V	2.0418	2.0241	0.9605	0.1007	0.9605

- Among the 15 biomarkers, pulmonary vascular FD showed the largest between-group difference: lower in early RILI. After FDR, FD had the lowest Padj.



- Early RILI had lower FD than non-RILI (2.192 vs 2.269; $P = 0.0055$), indicating reduced pulmonary vascular structural complexity.
- 3D reconstructions confirmed fewer distal branches and less spatial coverage in RILI (FD 1.9718 vs 2.3581).

CONCLUSION

We explored vascular-scale biomarkers to bridge whole-lung and pulmonary vascular scales: lower FD on routine non-contrast CT was associated with early RILI and may aid risk stratification, pending multicenter validation.